



# Active learning for cross-sentence n-ary relation extraction

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## ABSTRACT

N-ary relation extraction models are required to be trained on large amount of high-quality data, but it is challenging to obtain such data in effect; thus, models are forced to rely on a limited amount of low-quality labeled data. This paper proposes an active learning method for cross-sentence n-ary relation extraction, addressing the following question: “How can we train an n-ary relation extraction model incrementally without a large amount of high-quality data?” To answer this research question, we introduce a schema-aware sampling strategy that selects informative samples in unlabeled dataset that will be used for model training. This method exploits structural relatedness between a relation and its entities to generate the context embeddings of inferred relations. Using the similarity between the clusters of context embeddings and target samples, we detect a set of informative samples in unlabeled dataset. Moreover, the paper proposes a balanced incremental learning method updating the extraction model without bias with only a small computational cost at each training iteration. Experimental results on benchmark datasets are demonstrated to confirm the validity and effectiveness of the proposed method.

## 1. Introduction

Relation extraction is a promising approach to extract factual knowledge from unstructured text. Relation extraction aims to predict a relation from entities based on the context of sentences. The extracted relational facts now become an essential resource for intelligent applications such as question answering [1], recommender systems [2], and knowledge graph completion [3], etc.

Most existing relation extraction models focus on detecting a binary relation between two entities in a single sentence, known as intra-sentence relation extraction. However, relations can exist among more than two entities and appear across sentences. For example, as shown in Fig. 1, the *Victory* relation among *Roger Federer*, *2003*, and *Wimbledon Championships* entities appears in multiple consecutive sentences.<sup>1</sup>

A few recent works have tried to address a cross-sentence n-ary relation extraction problem [4–9]. Those methods employ graph-oriented neural network architectures such as graph-state LSTMs [6] and graph convolutional networks [10] to capture the context feature related to the n-ary fact. However, they suffer from a low-quality distantly-supervised dataset with numerous noisy facts, obstructing applicability to be used in real-world scenarios. It is challenging to fundamentally improve the extraction accuracy of the models unless sufficient labeled dataset is guaranteed. To address this research problem, this paper proposes an active learning

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<sup>1</sup> Examples are derived from the raw data in Wikidata.

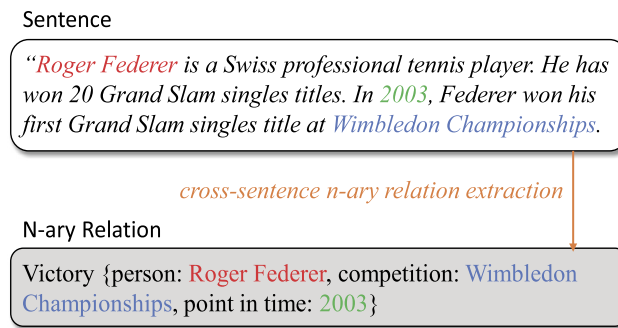


Fig. 1. An example of cross-sentence n-ary relation extraction. A Victory relation has three type-entity pairs.

method that allows researchers to train their relation extraction models while obtaining labeled data from unstructured text during model update.

Active learning has received attention for its capability to address such low-resource problems of deep neural models while retaining the strong learning performance in recent years. A typical active learning cycle is that a task model is trained on an initial labeled dataset. A sampling strategy finds informative samples from unlabeled data and hands them over to an oracle for their labeling. Finally, the queried samples are added to the labeled dataset and the task model is updated with the newly labeled dataset. This process is repeated until the label budget is exhausted or the pre-defined conditions are satisfied. Active learning is widely applied in various fields such as image classification and recognition [11–16], object detection [14], schema expansion [17], time series prediction [18], and regression [19,20] etc.

The key to active learning is to adopt a suitable sampling strategy to find informative unlabeled samples for the target task. Informative samples are those that improve the performance of models when annotated and added to the training dataset. In n-ary relation extraction, the structural relatedness between a relation and its entities is crucial. In other words, an n-ary relation extraction have to consider various combinations of argument entities and their semantic types. However, previous n-ary relation extraction methods cannot evaluate the validity of arguments and have difficulty in classifying an exact n-ary fact. For example, the Victory relation can be organized with only three types of argument entities including *person*, *competition* and *point in time*. However, previous methods do not care compatibility between the relation and its argument entity type. It means that they could extract the Victory relation containing entities of *food* or *vehicle* type, which is totally mistaken.

We introduce a schema-aware sampling strategy incorporating the topological information of n-ary relations with the data distribution in the entire training dataset. Specifically, the context embedding of inferred relations is generated from a relation schema embedding and a feature embedding of an n-ary relation extractor. Then, we build fine-grained clusters using this context embeddings and find the most informative samples by computing the similarities between clusters and the target samples.

In the active learning, the training model update should be carefully determined. A straightforward retraining method is to retrain the model from scratch using both existing and newly acquired data. However, it is clear that it requires much more time to train a model at each active learning iteration since the model has tremendous weight parameters as deep neural networks. Another option is fine-tuning on new data, but it is likely to be biased on the small amount of new data. To overcome the limitations, we propose a balanced incremental learning method which builds a training subset at each learning iteration consisting of new data and representative samples from clusters. Not only to relation extraction, but the proposed strategy can be adopted to various fields such as analyzing part-of-speech and learning language network. Such task highly depends on well-constructed data resources, and should take account of topological and semantic information [21,22].

To evaluate the proposed method, we compare the proposed sampling strategy with various sampling baselines in both simulation settings and human-in-the-loop settings. We also assess the effectiveness of the proposed training scheme compared with other training schemes. Additionally, the proposed active learning is compared with a non-active learning on training the entire models.

The major contributions of this paper are summarized as follows.

- The paper proposes an active learning method for a cross-sentence n-ary relation extraction (ANRE), which allows models to be trained on a small amount of labeled data initially. The proposed methods acquire newly labeled train data iteratively and improve the model.
- The paper proposes a schema-aware sampling strategy that finds informative samples using structural relatedness between relation and entities. A schema-aware sampling strategy incorporates the topological information of n-ary relations and the batch diversity.
- The paper introduces a balanced incremental learning to improve the models cost-effectively. The method builds a training subset consisting of newly labeled data and representative data from clusters at each active learning iteration.
- The paper compares the proposed sampling strategy with classic and state-of-the-art sampling methods in diverse experiment settings. We also assess the effectiveness of the proposed training scheme compared with different training schemes. Additionally, we analyze and demonstrate the impact of the components of proposed sampling strategy.

The rest of the paper is organized as follows. Section 2 describes related works and Section 3 introduces the proposed approach. Section 4 describes the experimental results and Section 5 concludes the paper.

## 2. Related work

The proposed method is primarily related to two strains of research: relation extraction and active learning. This section mainly focuses on recent knowledge extraction and active learning methods.

### 2.1. Relation extraction

Relation extraction is the task of extracting relationships between entity mentions. Most prior approaches rely on existing entity recognizers or entity typing systems. In this work, we focus on the models that integrate entity typing and relation extraction.

Augenstein et al. [23] introduce a learning method of jointly training the named entity classifiers and the relation extractor. However, it is dependent on existing NLP tools like entity recognizer. More recently, Yaghoobzadeh et al. [24] introduce multi-instance multi-label learning algorithms to perform entity typing and relation extraction without existing entity typing tools. CoType [25] runs a text segmentation algorithm to extract entities and jointly embeds entity mentions, relation mentions, and type hierarchy. Di et al. [26] introduce a method to extract a relation via domain-aware transfer learning (ReTrans). Takanobu et al. [27] propose a hierarchical reinforcement learning framework (HRL) for relation extraction. To extract overlapped and discrete relations, Zhang et al. [28] propose a multi-labeled relation extraction method that employs an attentive capsule network. The method finds low-level capsules that contain related relation features, and then clusters them to represent high-level features. Singh and Bhatia [29] define the relationships derived directly from a sentence as the first-order relation. The target relation is calculated by combining the first-order relation scores by referring to the indirect relationship connected by the context token as the second-order relation. Geng et al. propose a relation extraction model using sequential and tree-structured LSTM with attention [30].

Cross-sentence n-ary relation extraction is a recently explored task. Peng et al. [4] present graph-oriented long short-term memory networks for n-ary relation extraction. The method splits a document graph into two directed acyclic graphs (DAGs) and encodes each DAG using tree LSTM. Song et al. [6] propose a graph-state LSTM model (GSLSTM), which uses a recurrent state transition process to model rich graph linguistic contexts. GSLSTM incorporates sibling information into contextual vectors of entity mentions through dependency edges. Graph convolutional networks (GCN) are first adopted for intra-sentence binary relation extraction by [10]. Guo et al. [7] improve GCN model architectures for cross-sentence n-ary relation extraction. Jia et al. [8] propose a multiscale neural architecture for document-level n-ary relation extraction by expanding the extraction scope to the entire document. Ernst et al. [9] propose an approach that extracts fact candidates via distant supervision and exploit manual consistency rules to prune out false candidates.

### 2.2. Active learning

Active learning has been widely studied. Traditional active learning can be categorized as membership query-synthesizing methods, stream-based selective sampling methods, and pool-based methods [31]. Since pool-based query algorithm is more suitable for general application scenarios, recent works mainly focus on this direction.

#### 2.2.1. Uncertainty-based methods

The pool-based methods can be grouped into three categories: uncertainty-based methods, distribution-based methods, and hybrid methods combining uncertainty and distribution of query samples.

Uncertainty based methods estimate the uncertainties of samples and acquire top-K most informative samples. The classic methods of choosing samples are random sampling, least-confidence, margin sampling and entropy-based sampling. Random sampling randomly selects samples. Thus, it cannot judge which samples are the most informative. The least-confidence selects the least confident samples from the most likely labels using the task classifier's prediction. However, the sample chosen by the least-confidence is not always the most informative sample. If a sample uniformly shows predicted probabilities distributed over labels, the least-confidence cannot select the right sample. Margin sampling and entropy-based sampling consider the disadvantage of least confidence. They select the sample that has the most uncertain probability distribution over the two most probable labels or all labels. However, those methods also depend on unreliable task model's prediction.

Since classic approaches do not scale well to deep neural networks, some works rely on Bayesian networks [32,33,15,34]. Houlisby et al. [32] propose the Bayesian active learning by disagreement scheme (BALD) that measures the mutual information between model parameters and the model predictions. However, BALD is likely to choose similar samples, which wastes labeling resources and makes it difficult for the model to learn useful information. Gal et al. [15] present the DBAL, which utilizes Monte Carlo dropout methods to combine Bayesian neural networks with BALD. However, recent works [35] point out that DBAL is not scalable for large datasets due to batch sampling problem, and also likely to be fooled by adversarial examples. Tran et al. [34] introduce the Bayesian generative active deep learning method to generate informative samples for training. This method aims to generate useful batch samples of different categories. BatchBALD [33] expands the idea of BALD [32] to the batch query scenario. It acquires multiple informative samples by approximating the mutual information between a batch of samples and model parameters.

In non-bayesian methods, Beluch et al. [36] propose the ensemble-based sampling method using the Monte Carlo Dropout. AL-MV [37] employs multiple views from hidden layers of CNN and measure the uncertainty. AL-MV algorithm is introduced to perform

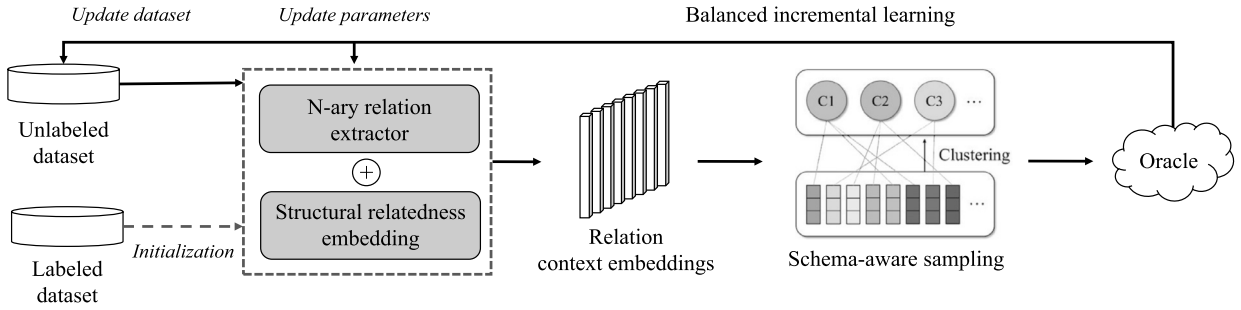


Fig. 2. A proposed active learning approach for cross-sentence n-ary relation extraction.

adaptive weighting of the uncertainty of each layer. Yoo et al. [14] introduce a task-independent loss prediction module for active learning. The loss prediction module predicts the loss of the task module, and then top-K predicted losses are selected as uncertain samples. Although the uncertainty-based methods perform well on various tasks, they cannot fully reflect the diversity of batch samples. Also, the sampling performance of these methods tends to decrease as the number of target labels increases.

### 2.2.2. Distribution-based methods

Distribution-based methods choose uncertain samples based on the distribution of extracted model features. These methods try to construct the compressed dataset to represent the entire dataset's feature space. Chitta et al. [38] use an ensemble active learning instead of random sampling to build a training subset based on data distribution. They prove that active learning with the compressed dataset may show better performance in the large-scale sampling scenario. Wang et al. [39] introduce cost-sensitive active learning with statistical methods algorithm. They compute the optimal number of query labels using hypothetical distribution models. The Core-set approach [35] constructs a core subset representing the remaining unlabeled data samples. This approach provides a bound between an average loss over given subsets of the dataset and the remaining data points. The Core-set approach shows that minimizing the bound is equivalent to solving the k-center problem. However, this method requires a large distance matrix, computationally expensive in the active learning scenario.

We confirm the major drawback of distribution-based methods that the target task pipeline cannot be considered. Also, these methods cause selecting overly representing data points, since they try to construct the subset using data points as far as possible. Moreover, these approaches do not take account how outputs depend on input data.

### 2.2.3. Hybrid methods

To combine uncertainty and the distribution of data samples, BADGE [40] exploits gradient embedding and k-means++ seeding algorithm. It embeds the unlabeled data points and then clusters the embedded data points. However, BADGE regards the model prediction of the unlabeled data as gold truth to calculate the gradient loss. Meanwhile, VAAL [12] learns the uncertainty and distribution of data using VAE and adversarial networks. VAAL uses both labeled and unlabeled datasets to train the generative adversarial network in a semi-supervised manner. TA-VAAL [13] expands the VAAL architecture. TA-VAAL integrates the loss prediction module and RankCGAN [41] with the VAAL architecture to elaborately consider model uncertainty and data distribution. SRAAL [11] is a state re-labeling adversarial active learning framework. A representation generator transforms the annotation information into the representations via the supervised target learner and unsupervised image reconstructor. Then, the online uncertainty indicator re-label the state of unlabeled samples, and the state discriminator is updated using state losses. With the rise of a large language model, there are a few works exploring the possibility of utilizing those models with active learning paradigm. ALPS [42] uses the masked language model loss derived from a pre-trained language model to classify uncertain samples in cold-start settings. BATL [43] introduces an active learning with the triplet loss which uses loss extracted from a pretrained language model to distinguish informative samples with similar features

However, current hybrid methods also have some limitations. BADGE is significantly dependent on unreliable confidence scores of task models. VAAL-based methods require VAE frameworks and use the pair-wise loss focusing on loss value itself or relative ranking score.

## 3. Schema-aware sampling with balanced incremental learning

The proposed approach consists of three components: (1) cross-sentence n-ary relation extractor for generating context embeddings of the n-ary relation, (2) schema-aware sampling for selecting the most informative data samples to annotate, and (3) balanced incremental learning for efficiently updating model parameters using newly annotated data samples and representative data chosen from trained data samples. The overall architecture is shown in Fig. 2.

### 3.1. Preliminaries

We first demonstrate the concept of n-ary relations and their representations and then formulate the n-ary relation extraction problem. N-ary relation extraction is the task that predicting semantic relations  $r \in R$  among multiple entities expressed by text

given typed entities  $e \in E$  in a raw text consisting of consecutive sentences. We represent an n-ary relation fact in the form  $r(type_1 : e_1, \dots, type_n : e_n)$  following [44] where  $type$  is the semantic type of argument entities, and  $e_1$  through  $e_n$  are the argument entities of a relation  $r$ . Based on the concepts above, there is a schema of n-ary relation to formally define the entity types that the relation type can have as arguments. An n-ary relation fact consists of pairs of an entity type and an entity. For example, when given a text that “Roger Federer won Wimbledon Championships in 2003” and an n-ary relation schema  $Victory\{person \times competition \times pointintime\}$ , the extracted n-ary relation has three arities and consists of three type-entity pairs as follows:

$$\begin{aligned} r &= \{type_1 : e_1, \dots, type_n : e_n\}, \\ \rightarrow Victory &= \{person : RogerFederer, \\ &\quad competition : Wimbledon \\ &\quad Championships, \\ &\quad pointintime : 2003\} \end{aligned}$$

### 3.2. Relation context embedding

Our approach is not restricted to specific extraction models. As we can leverage any state-of-the-art relation extraction models, we mainly utilized the graph-state LSTM model presented by [6] in this work. The brief description is as follows. First, input sentences are parsed to construct an input graph with dependency edges. GSLSTM uses a recurrent state transition process, allowing word-level sub-states to exchange information through dependency edges. It incrementally refines a neural graph state value and reflects structural information in n-ary relation extraction tasks. Before classifying the relation types with multiclass classifier, GSLSTM generates hidden states containing the syntactic and semantic context of input sentences and entities.

$$h_i = GSLSTM(h_{i-1}, \tilde{\theta}), \quad (1)$$

where  $h_i$  is the  $i$ -th hidden state and  $\tilde{\theta}$  is the set of trainable parameters. The model parameters are updated via a cross-entropy loss  $L_{extractor}$  over a set of ground-truth labels.

We aggregate the hidden states to generate relation context embeddings, which are used for data sampling. Each hidden states contain the context of syntactic and semantic information of sentences. However, we do not assure the validity of an n-ary relation with the aggregated information alone from an n-ary extractor. In other words, we need to consider whether the type-entity pairs form a valid relational fact or not.

To tackle the problem, we incorporate the structural relatedness of all the type-entity pairs into the relation context embedding. A schema information is encoded via the NaLP method [44] in order to inject the different topological information of n-ary relation schema in relation context embedding. We first compute the type-entity pair embedding  $M_r$  through convolution and concatenation process. Then, the relatedness score  $s_r$  between the relation  $r$  and the argument entities is evaluated via a fully connected network. A relatedness score will be higher if the type-entity pairs are expected to form a valid relation. When capturing the relatedness of a set of type-entity pairs, we apply element-wise minimizing over the pair-wise relatedness for any two type-entity pairs to approximately evaluate their overall relatedness, since the minimum relatedness of two type-entity pairs of a valid n-ary relation need to be sufficiently large. The relatedness score  $s_r$  is defined as follows:

$$s_r = FCN(concat(conv(M_r))), \quad (2)$$

where  $s_r$  is a relatedness score,  $M_r$  is the embedding matrix of type-entity pairs,  $conv$  is the convolution operation,  $concat$  is the concatenation operation, and  $FCN$  is the fully connected network with  $\tanh$  activation function.

Through evaluating the relatedness score of relations, we can incorporate the structural information of relations in a relation context embedding. According to the relatedness score, we update the type-entity pair embedding  $M_r$  through the loss function  $L_{relatedness}$  as follows [45]:

$$L_{relatedness} = \log(1 + e^{-I_{M_r} s_r}), \quad (3)$$

where,  $I_{M_r} = 1$ , if  $M_r$  is a valid relational fact, and  $I_{M_r} = -1$ , otherwise.

With the aggregated hidden features from n-ary relation extractor and updated type-entity pair embeddings, we finally define the relation context embedding  $c_r$  as follows:

$$a_i = f(M_r \cdot W_a \cdot h_i) \quad (4)$$

$$c_r = \sum a_i h_i, \quad (5)$$

where  $a_i$  is the attention of structural information,  $f()$  is a softmax function which normalize the attention weights,  $W_a$  is the trainable parameter matrix, and  $h_i$  is the  $i$ -th hidden state of the relation extractor. Dimension  $d$  for each variable is as follows:  $d_{h_i} = 200$ ,  $d_{a_i} = 200$ ,  $d_{M_r} = 200 \times 200$ , and  $d_{W_a} = 200 \times 200$ .

**Algorithm 1** Active learning for n-ary relation extraction.

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**Input:** unlabeled text  $D^u$ , labeled text  $D^l$ , n-ary relation extractor  $\mathcal{M}$ , sampling size  $k$ , iteration number  $\mathcal{T}$ ;  
**Output:** n-ary relations sampled at  $i$ -th iteration  $\mathcal{T}_i$ ;  
1: initialize  $\mathcal{M}$  on  $D^l$ , generate clusters  $\mathcal{G}$  using relation context embeddings  $c_i$  in  $D^l$ ;  
2: **while**  $\mathcal{T}_i$  in  $\mathcal{T}$  **do**  
3:   generate relation context embeddings  $c^u$  on unlabeled text  $D^u$ ;  
4:   compute entropy scores  $H$  of  $c^u$  over  $\mathcal{G}$ ;  
5:   select  $k$  samples in order of high  $H$ ;  
6:   query oracles to annotate  $k$  samples;  
7:   receive newly labeled data  $D^{new}$  according to feedback;  
8:    $D^l \leftarrow D^l \cup D^{new}$ ,  $D_u \leftarrow D^u \setminus D^{new}$ ;  
9:   build training subsets using  $D^{new}$  and  $n$  representative samples per clusters  $\mathcal{G}$ ;  
10:   update model parameters;  
11:   update clusters  $\mathcal{G}$  using  $D^l$ ;

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The relation context embedding contains an uncertainty about a sample and is updated via the final loss defined as follows:

$$L_{final} = L_{extractor} + \lambda \cdot L_{relatedness}, \quad (6)$$

where  $\lambda$  is a scaling parameter.

### 3.3. Schema-aware sampling strategy

A crucial part of a good sampling strategy is developing an effective measure of informativeness suited for a target task. For example, in this task, if we use the confidence of relation context embeddings as an exclusive measure of informativeness, it is likely to miss the information of data distribution in a batch of samples. In other words, the sampling strategy for relation extraction should consider a various information in the relational facts including semantic context of fact and diversity of data samples.

Considering the above challenges, we generate clusters based on the relation context embeddings to distinguish samples according to their similarities. We adopt the k-means++ clustering algorithm [46] to calculate the centroids sequentially. The centroid is selected in proportion to the square of the minimum distance between the previously selected centroids and other possible candidates.

To find the optimal number of clusters, we use the average silhouette method [47]. The silhouette value represents how similar a sample is to its own cluster when compared to other clusters. From the -1 to +1 range, a higher silhouette value indicates that the sample belongs to the appropriate cluster. A silhouette value  $S_i$  is calculated as follows:

$$S_i = \frac{b(i) - a(i)}{\max(a(i), b(i))}, \quad (7)$$

where  $a(i)$  is the average distance between a data sample  $i$  and other data samples within the same cluster, and  $b(i)$  is the average distance between a data sample  $i$  and other data points in the closest cluster. We can find the optimal number of clusters when the average silhouette value of all clusters exceeds the threshold.

Since the optimal number of clusters can be higher than the number of relation types, we can divide samples of the same relation type into several clusters according to the context in the relational facts. For example, relations representing *Victory* in sports events can form a single cluster, whereas relations representing *Victory* in elections form the other cluster, despite having the same relation type.

When unlabeled text is injected, our proposed method generates relation context embeddings. Then we calculate the Shannon entropy scores  $E$  using the relation context embedding as variables and the generated clusters as classes as follows:

$$E = - \sum_i \sum_j P(g_j | v_i) * \log_2 P(g_j | v_i), \quad (8)$$

where  $g_j$  indicates  $j$ -th cluster, and  $v_i$  indicates  $i$ -th relation context embedding.  $P(g_j | v_i)$  is obtained by cosine similarity between the center of cluster  $g_j$  and the relation context embedding  $v_i$ .

The entropy scores reflect the probability that a chosen sample belongs to each cluster, and the most ambiguous sample would have the highest scores. Based on the entropy scores,  $k$  most ambiguous samples are selected to query oracles. After the annotation is done, the newly labeled data are used to update the model and clusters.

### 3.4. Balanced incremental learning

The most intuitive way to update the model is to retrain the entire model from scratch using both the newly labeled and existing data. However, it is difficult to apply a retraining scheme in practice since it is time-consuming and costly. Fine-tuning the model with newly labeled data might be considered as another solution, but it is likely to bias the model on a small amount of new data. Moreover, fine-tuning has more problems with a large number of target labels such as in relation extraction task. As studied in [48], there are two factors: 1) data imbalance between the old and new labels, and 2) the increasing number of similar labels. Thus, we need to update a model in a different way, considering the above deficiency of learning methods.

We propose a balanced incremental learning method of using newly labeled data with a subset of old data which is previously used for training. To build a subset of training data, a fixed number of representative samples are extracted from each cluster. We



compute the distances of samples to the center of the cluster, assuming the samples close to the center of a cluster have the most distinguishable feature. Samples that are closed to the center of clusters are then selected and served as representative samples in training subset for model update. The best ratio of new data and old data for a balanced training subset is discussed in the experiment section.

## 4. Experiments

This section describes the effectiveness of the proposed approach through various comparative experiments.

### 4.1. Experimental settings

#### 4.1.1. Datasets

We used three publicly accessible datasets in the experiments.

- PubMed: The dataset introduced in [4], a biomedical-domain dataset focusing on drug-gene-mutation relations, containing 6987 ternary instances extracted from PubMed. Most instances include multiple sentences where the average number is around 2. There are five relation types: “resistance or nonresponse”, “sensitivity”, “response”, “resistance” and “None”.
- DocRED: DocRED was introduced in [49]. It consists of binary relations extracted from 5,053 Wikipedia documents. It contains 1,508,320 and 38,269 instances in training set for weakly supervised and supervised setting, 12,332 instances in the dev set, and 12,842 instances in the test set. There are six types of entities, including ‘person’ and 96 types of relations, such as ‘place of birth’ and ‘award received’. We divided DocRED into DocRED-weak (weakly supervised training set) and DocRED-sv (supervised training set) and used them for the experiments.
- Wiki-KBP: Wiki-KBP consists of 23,884 training sentences sampled from Wikipedia articles. 289 sentences were sampled from the 2013 KBP slot filling assessment results for the test data. It is annotated with 126 entity types and 13 relation types. We use the preprocessed Wiki-KBP dataset introduced in [25]. Since the test data was generated via distant supervision, we provide an approximate measure of performance.

Although DocRED and Wiki-KBP do not have high-arity relations, they have more sentences per instance and diverse entity and relation types than PubMed. Different characteristics of the three datasets help to investigate the impact of the arity of relations, the lengths of consecutive sentences, and the diversity of relation types.

#### 4.1.2. Baselines

We compared our sampling strategy with the following strategies:

- RAND (random sampling): selects random  $k$  samples
- CONF (least confidence sampling): selects  $k$  samples with the least confident labels inferred by the classifier [50]
- MARGIN (margin sampling): selects  $k$  samples with minimum margin between two most confident labels inferred by the classifier [50]
- ENTROPY: selects  $k$  samples with highest Shannon entropy inferred by the classifier [50]
- CORESET: selects  $k$  samples using core subset selection with a greedy furthest-first traversal conditioned on all labeled samples [35]
- BADGE: selects  $k$  samples based on the gradient loss of classifier and  $k$ -means++ clustering [40]

#### 4.1.3. Training scheme

We compared our training method with the following training schemes for active learning: Retrain (RT) trains the model on the entire labeled data from scratch at each iteration. Fine-tune(FT) directly fine-tunes the previous model using the newly labeled data.

#### 4.1.4. Setup

In simulation experiments on three dataset, we initialized the  $n$ -ary relation extractor with about 20% of training data. We used the remaining 80% of training data as unlabeled data and annotated 5% of the unlabeled data for each iteration using gold labels. We evaluated the model’s prediction accuracy using test data.

We also conducted human-in-the-loop experiments for the DocRED-weak dataset. Five annotators of varying annotation proficiency participated in the experiments and had enough time to get used to the dataset. Each participant annotated about 100 samples for 10 minutes at each iteration. To compare the results under the same conditions as in simulation experiments, we annotated the remaining samples assigned to each iteration by using gold labels. Finally, we reported the average of experimental results of five annotators. In these comparative experiments on sampling strategies, we trained the models using a balanced incremental learning method at each active learning phase.

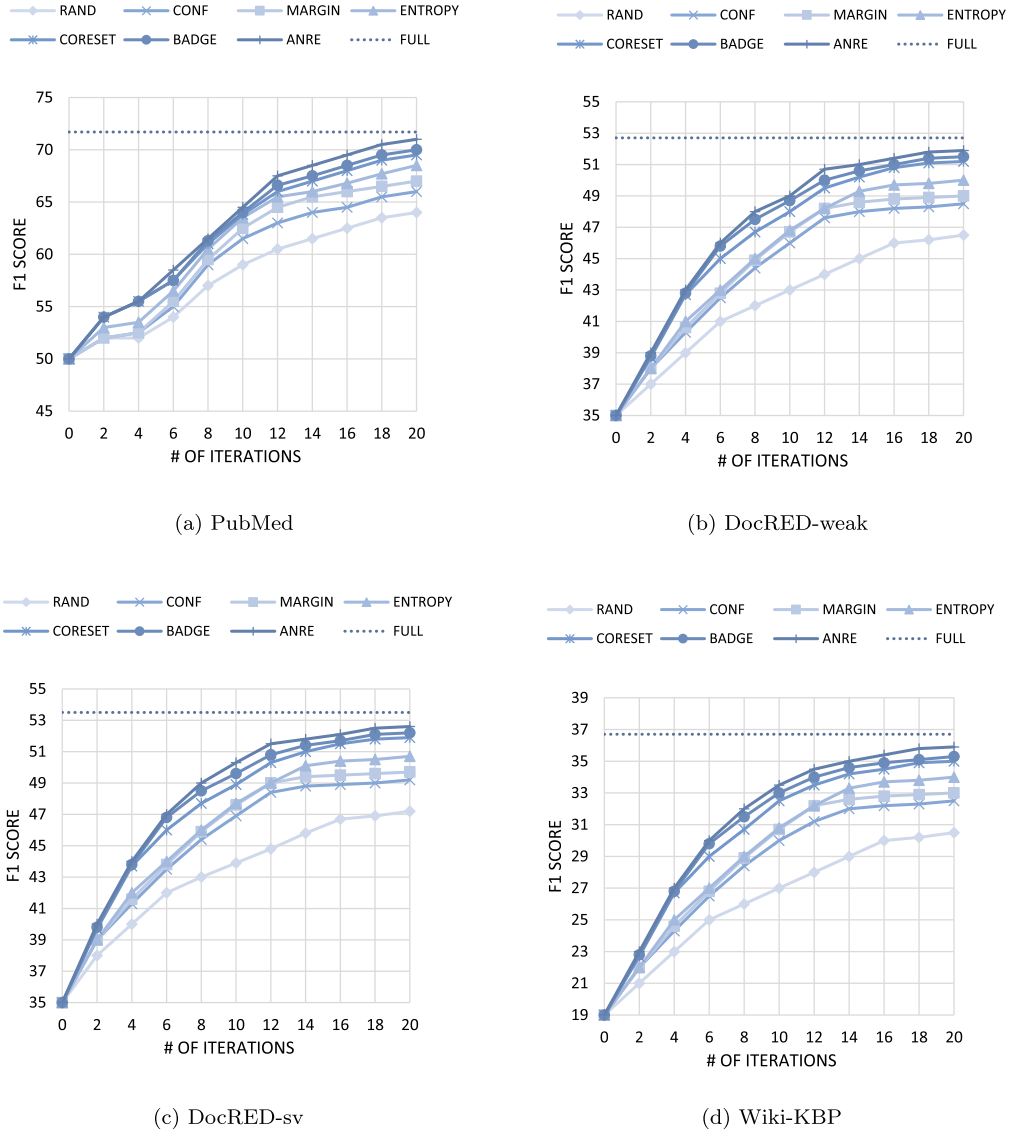


Fig. 3. Comparison of sampling strategies on four datasets: F1-scores at each active learning iteration in the simulation setting.

## 4.2. Comparison of sampling strategy

### 4.2.1. Results

From Fig. 3, we found that other sampling strategies outperform random sampling in every experimental setting. It demonstrates that the specific informativeness measure need to be set up according to tasks. Least-confidence also shows relatively low prediction accuracy, which indicates that relying on the least-confident label of samples does not help with finding informative samples. Similarly, we observe that margin sampling has about 1% higher accuracy than the least-confidence. Entropy has the highest accuracy among traditional sampling strategies. However, it falls behind our proposed method by about 3%. While entropy simply calculates the probability distribution over relation labels, e.g., “sensitivity” as a class label, the proposed method adaptively generates clusters and then calculates the probability distribution over clusters.

Moreover, ANRE performs better than state-of-the-art baselines. ANRE achieves about 1.5% higher accuracy than CORESET. While our method considers both the uncertainty of extracted n-ary relations and the diversity of selected samples, CORESET only considers the distribution of samples. Also, ANRE slightly performs better than BADGE by about 1%. BADGE finds task-related uncertain samples using gradient loss of an n-ary relation extractor. It is impressive that BADGE shows similar F1 score even without structural information of n-ary relation. It indicates that gradient loss could be one of the most important factors for measuring the uncertainty of n-ary relations. It is evident that exploiting the context of sentences and structural features of relations plays a positive



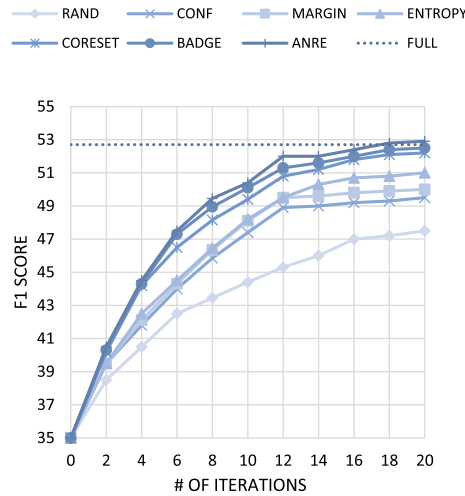


Fig. 4. Comparison of sampling strategies on DocRED dataset: F1-scores at each active learning iteration on DocRED-weak in the human-in-the-loop setting.

role in selecting informative samples for relation extraction, but there's still bags of room for improvement. In the future work, we are planning to improve ANRE using gradient loss and other related factors when we compute the relation context embedding.

#### 4.2.2. Impact of human annotation

Compared with the simulation, human annotation improves the model's performance in the experiments on the weakly-supervised training dataset. As shown in Fig. 4, F1-scores in human-in-the-loop settings are about 2% higher than those in simulation settings in the early training iterations and are 1% higher at the convergence. Like the simulation experiments, our proposed sampling strategy performs better than baseline methods. Since the weakly-supervised training dataset contains noisy relations, the oracle could correct those incorrect relation labels. However, there is an obvious tradeoff between performance improvement and additional time and cost for human resources. Moreover, in real-world scenario, we assume that human annotation could lead to produce varying experimental results depending on how a system user aggregates and utilizes each human assessment.

#### 4.2.3. Impact of the relation schema

PubMed consists of ternary relations, while DocRED and Wiki-KBP are composed of binary relations. From the experimental results on three datasets, our proposed method improved the inference accuracy at every training step regardless of the arity of relations. Meanwhile, the performance gain on DocRED and Wiki-KBP were much lower than that on PubMed, which indicates that the number of relation types and entity types has a significant impact on the model's performance. When the models initialized with 20% of the training data, F1-score on PubMed is about 50%. When the proposed model was learned with 100% of the training data in simulated active learning settings, F1-scores on PubMed is about 71%, which shows 21% performance gain. On the other hand, performance gains in the DocRED-weak and DocRED-sv in simulation settings are 17% and 17.5%, respectively. Considering that the supervised DocRED training set has slightly more accurate data than weakly-supervised set, we also confirm that the quality of the training data influences the model's performance as shown in Fig. 3(b) and 3(c).

#### 4.2.4. Impact of the number of clusters

We analyzed the impact of the number of clusters on the model's performance. In this experiment, we fixed the number of representative samples for each of the clusters in the balanced incremental learning. Fig. 5 show the experimental results on each dataset. The results demonstrate that it is crucial to set the appropriate number of clusters. We found that the optimal number of clusters varies for different dataset. We observed that the influence of structural context of inferred relations increases in large number of clusters, but the computational cost also increases. We found the optimal number of clusters for each dataset through the average silhouette method empirically:  $n = 10$  on the PubMed,  $n = 1000$  on the DocRED-weak, and  $n = 100$  on the Wiki-KBP dataset. Although we could find out the optimal number of clusters, it requires additional time for computing average silhouette width score. In the future work, we are planning to design an algorithm which can adaptively find the optimal number of clusters. We expect that it leads to accelerate the efficacy of an active learning, especially for capturing diversity of batch samples.

#### 4.2.5. Comparison of training scheme

We investigated the performance for each training scheme as shown in Figs. 6 and 7. As we expected, RT showed the best performance in every experiment. Since RT uses all collected data to train the model from scratch, it had a positive impact on the prediction accuracy. FT shows about 1% lower F1-scores than the proposed method. The proposed method shows almost similar performance to retrain, which indicates that we can efficiently train the model without bias.

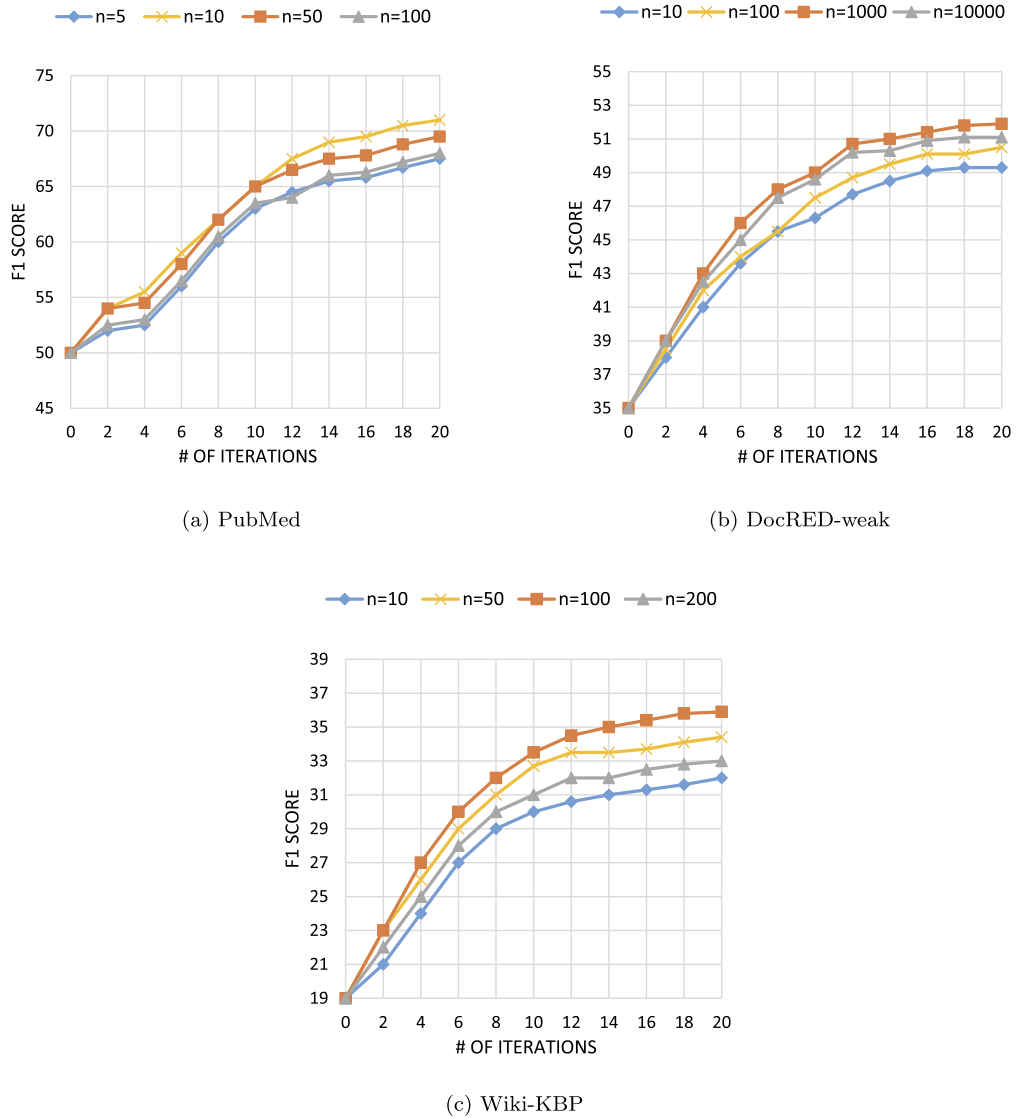


Fig. 5. Impact of the number of clusters: F1-scores at each active learning iteration on three dataset.

Table 1

Average model training time per iteration (in seconds). 20% of training data is used for initialization. 3% of the training data is used at each iteration in simulation settings.

|                   | PubMed | DocRED-weak | DocRED-sv | Wiki-KBP |
|-------------------|--------|-------------|-----------|----------|
| non-active (40%)  | 988    | 71,400      | 2,787     | 2,234    |
| non-active (100%) | 2,574  | 187,625     | 8,458     | 6,480    |
| retrain           | 1,881  | 126,925     | 5,936     | 4,302    |
| fine-tune         | 78     | 5,095       | 207       | 156      |
| balanced IL       | 83     | 5,472       | 254       | 243      |

#### 4.2.6. Time efficiency

To evaluate the model training efficiency of the balanced incremental learning method in the simulation settings, we first initialized the models using 20% of training data. We used 3% of the training data at each iteration to estimate the average training time per iteration. Table 1 reports that the balanced incremental learning requires much shorter training time on every dataset. It indicates that although RT shows slightly higher F1-scores for relation extraction, our proposed training method is much more suitable for active learning since RT requires 20 times more training time. We also confirmed that since it takes additional time for the balanced incremental learning method to build training subsets, FT shows slightly shorter training time than the proposed method.

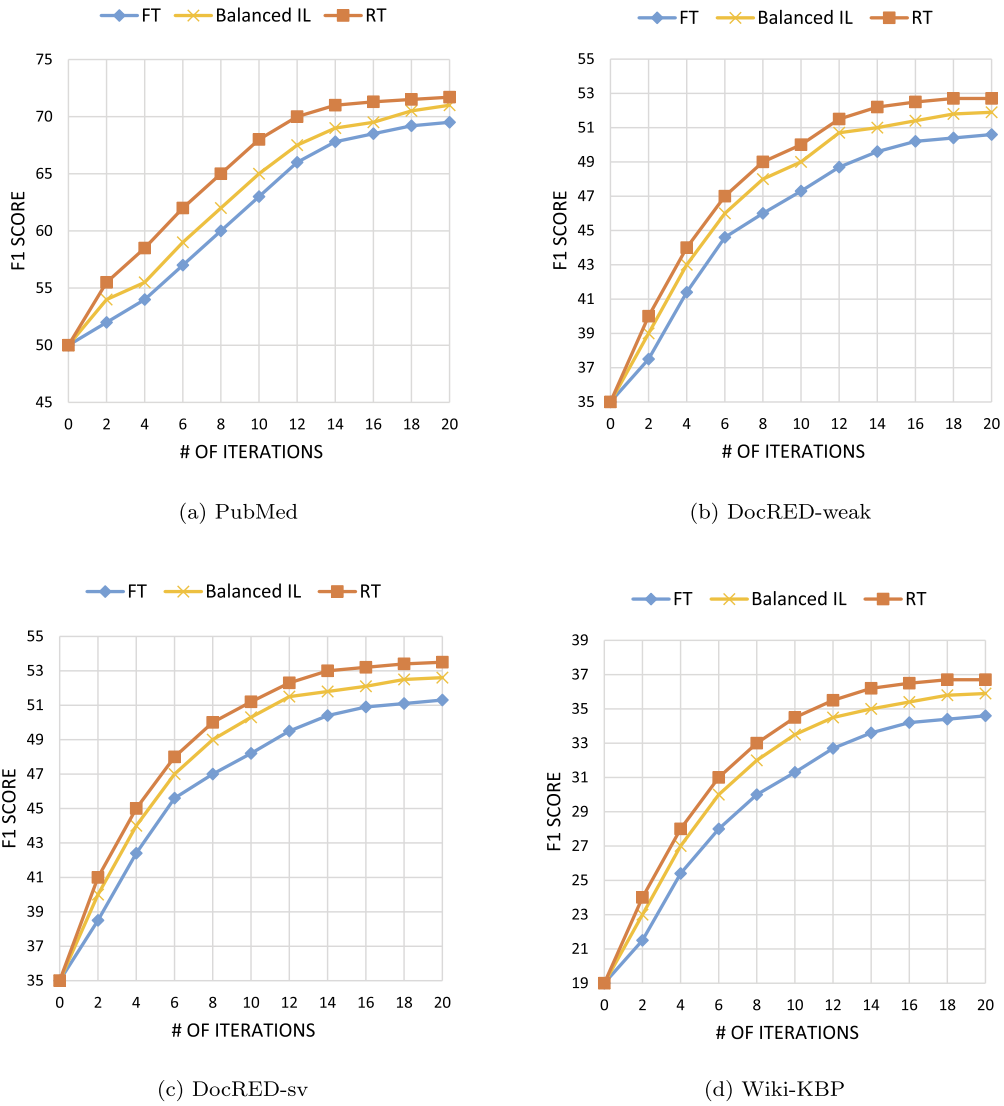


Fig. 6. Comparison of training schemes on four dataset.: F1-scores at each active learning iteration in the simulation setting.

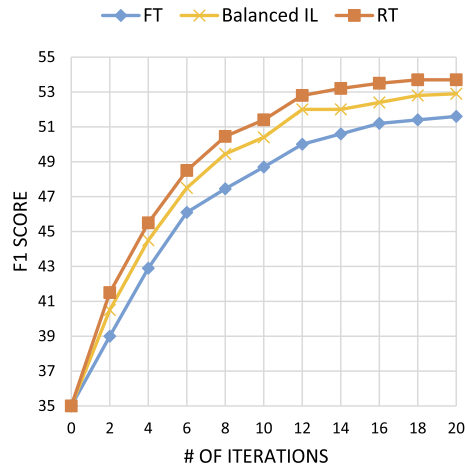


Fig. 7. Comparison of training schemes on DocRED dataset: F1-scores at each active learning iteration on DocRED-weak in the human-in-the-loop setting.

**Table 2**  
Impact of the number of representative samples per cluster.

|                    | PubMed |      |     | DocRED-weak |      |      | Wiki-KBP |     |     |
|--------------------|--------|------|-----|-------------|------|------|----------|-----|-----|
| # relation types   | 5      |      |     | 96          |      |      | 13       |     |     |
| # samples/clusters | 50     | 75   | 100 | 100         | 200  | 300  | 100      | 200 | 300 |
| F1-Score (ANRE)    | 70.2   | 70.6 | 71  | 51.3        | 51.7 | 51.9 | 35.9     | 36  | 36  |

**Table 3**  
Comparison of n-ary relation extractor on three dataset. F1-Scores after 20 active learning iteration.

| N-ary Extractor | PubMed | DocRED-weak | Wiki-KBP |
|-----------------|--------|-------------|----------|
| GSLSTM          | 71.0   | 51.9        | 35.9     |
| GCN             | 74.5   | 55.6        | 37.3     |
| AGGCN           | 76.8   | 56.2        | 3.7      |

#### 4.2.7. Impact of the number of representative samples

We also compared the impact of the number of representative samples per cluster. We evaluated the model performance with the different number of samples per clusters over three dataset: 50, 75, and 100 for the PubMed, and 100, 200, and 300 for the DocRED-weak and Wiki-KBP. We measured the F1-score in the simulation active learning settings with 100% of training data. The number of clusters was fixed to the optimal values:  $n = 10$  on the PubMed,  $n = 1000$  on the DocRED-weak,  $n = 100$  on the Wiki-KBP.

Table 2 shows the results of these experiments. As the number of representative samples per cluster increases, the better the accuracy obtained in both datasets. For 50 samples per cluster in PubMed, the F1-score is about 1% lower than that of 100 samples per cluster. Compared with the F1-Score in the FT training scheme for the PubMed dataset (69.5%), ANRE using balanced incremental learning achieves 1% higher accuracy even with 50 samples per cluster setting. We confirmed that using representative samples helps the model to avoid the bias of newly obtained samples.

#### 4.2.8. Comparison of n-ary extractors

We analyzed the impact of n-ary relation extractor on the proposed approach. In this experiment we use three different n-ary relation extractors, GSLSTM, GCN [5], and AGGCN [7]. GCN employs graph convolutional networks model over a pruned tree to extract n-ary relations, and AGGCN is an attention guided graph convolutional network model which adopts soft pruning strategy to learn a graph representation. Table 3 reports the experimental results. Although we mainly described the GSLSTM as an n-ary relation extractor of the proposed approach and the other extractors achieve the higher performance, this experiment shows that the proposed approach is flexible and can be extensively applied to various relation extraction models.

## 5. Conclusion

This paper proposed an active learning method for n-ary relation extraction. The proposed method allows an initial relation extraction model to be trained on a small amount of labeled data. The model is then updated with the newly labeled data at each active learning iteration. A schema-aware sampling strategy incorporates structural information of n-ary relations and batch diversity of given samples. Moreover, we proposed a balance incremental learning method that helps models to be updated cost-effectively without bias, where a training subset is adaptively built at each learning iteration.

We conducted extensive experiments on two benchmark datasets. We compared the proposed method with state-of-the-art baselines, and confirmed that the proposed method achieved state-of-the-art performance. We also analyzed the effectiveness of balanced incremental learning and the impact of each component of proposed sampling strategy.

Although we showed that the proposed method is useful for relation extraction, we need to improve our method in order to apply to more general application scenarios like complex event processing. We plan to investigate the most important uncertainty measure for active learning, which can be applied to variety of NLP tasks.

## CRedit authorship contribution statement

**Seungmin Seo:** Conceptualization, Methodology, Software, Writing – original draft. **Byungkook Oh:** Data curation, Formal analysis. **Jeongbeom Jeoung:** Software, Validation. **Donghyun Kim:** Software, Validation. **Kyong-Ho Lee:** Project administration, Writing – review & editing. **Dong-Hoon Shin:** Conceptualization, Funding acquisition. **Yeonsoo Lee:** Conceptualization, Funding acquisition.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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